Kacheong Li (Jacky)

Software Development Engineer
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Technical Skills

Coding: C++, C#, Python | **Tools:** Perforce, Visual Studio, ImGUI, Jira, AWS

Professional Experience

Engine Programmer ***Feb. 2021 - Present***
Cat Daddy Games (A Take-Two Interactive Studio)
Automation & Cloud Tooling

- Optimized game launch asset delivery pipeline by implementing **hash-based DLC caching and segmented asset loading**; enabled clients to download only missing or outdated assets and defer non-critical content to background downloads during gameplay, reducing first-time user experience (FTUE) wait time from **~2 minutes to ~10 seconds across 10M+ player mobile titles**.
- Developed a Python-based automated video encoding and deployment pipeline using **FFmpeg and AWS S3**, enabling producer self-service publishing of **rewarded advertisement content** (upsell/cross-sell placements via mediation platforms such as **IronSource**) with adaptive bitrate HLS streaming.
- Reduced deployment effort by **80–95%**, eliminated manual engineering bottlenecks, and saved an estimated **$5K–$10K annually** in operational and bandwidth costs.

NBA 2K Mobile – Rewind & Highlight Mode (Season 7)

- Contributed client-side development of a live NBA highlight–driven game mode using **C#/.NET**, serving a **5M+ players** base.
- Implemented server-driven, data-driven architecture to generate dynamic gameplay objectives from real-world NBA statistical events.
- Increased player participation by **~40–60% compared to prior live event modes** during launch windows, with **~65–70% conversion** from entry to completion.
- Built a **UWP-based internal live data editor** enabling LiveOps teams to configure and deploy content without client updates.
- Instrumented **analytics and telemetry pipelines** to track user behavior, optimize engagement funnels, and support data-driven iteration.

Catnip – Real-Time 3D Creation Platform (C++ / ImGui)

- Internal tooling platform powering three globally released 2K mobile live-service titles generating **$30M+ in annual revenue** and reaching **10M+ of lifetime downloads**.
- Refactored internal design editor using MVVM architecture and modern IMGUI interface to speed up scripting animations and game scene from hours to minutes
- Architected a data-driven blueprint layer and in-editor asset manipulation system that eliminated recursive 3D tool → runtime iteration cycles and fragile alias-based bindings, reducing asset validation time from **~1–3 hours to ~10–20 minutes** (~70–85%), and reducing scene initialization code from **hundreds of lines to a single** blueprint-driven call (~60–80%), enabling designers to modify assets without runtime code changes.
- Built an **automated UI testing** and onboarding framework for Catnip using a customized ImGui test library, simulating real user inputs and introducing “story”-driven end-to-end workflows; enabled fast, deterministic regression testing while repurposing test scenarios as **interactive tutorials** for user onboarding, reducing manual validation effort and improving usability.
- Redesigned **Perforce API** integration layer, improving sync reliability and conflict detection; kept manual submit conflict resolution **below 10%** of cases, significantly lowering cross-team integration friction.
- Drove platform evolution from prototype to v2.0 beta, implementing version-aware scene loading and backward-compatible data migration to maintain uninterrupted live-service workflows during upgrades.
- Refactored logging and notification framework using the Decorator pattern, enabling structured logging and **actionable error diagnostics**; reduced debugging turnaround time across teams.
- Mentored and onboarded 3 engineers via documentation, architecture walkthroughs, and code reviews; accelerated v2.0 feature delivery to beta within two months.

Academic Projects

Physics, Architecture Programmer & Art Designer ***Sep. 2019 – Aug. 2020***
Re: Chromatic, C++ (4 Members)

- Implemented ECS game engine architecture with custom memory allocator and thread pool in C++
- Architected OOBB 3D physics system capable of handling over 1k collisions with over 60fps
- Prototyped boat-themed endless runner with Python
- Modeled and animated boss and ship models with IK rigging using Blender

Solo Contributor ***Jan. 2020 – Apr. 2020***
Multiple Rigid Body Collision Simulation, C++

- Constructed BVH tree using surface area of AABBs as heuristic
- Implemented ISA-GJK and SAT collision detection for narrow phase
- Developed sequential impulse contact solver with split impulses approach for collision resolution
- Simulated motion and collision of over 300 stacked boxes together with performance of over 25fps

- Rendered scene and debug wireframe using shaders and NanoGUI

Gameplay Programmer & Art Designer ***Feb. 2019 – Apr. 2019***

Skyward, C++ (3 Members)

- Implemented component-based game engine architecture and modified 2D physics engine with depth correction during camera rotation
- Optimized collision and storage of voxel terrain from $O(n^3)$ to O(nlogn).
- Utilized customized level editor to design nine levels containing different visualization features and multiple paths for solving puzzles

Vibe Coding

Ray Tracing Simulation, C++

- Simulated light path tracing with reflection BRDF, transmission, image-based lighting, and depth of field
- Optimized speed to 1.035 sec to generate 400*300 pixels every pass
- Developed a progressive ray tracing engine with a CPU reference renderer using existing Eigen-based BVH acceleration.
- Implemented scene parsing, HDR image output, and CPU/GPU rendering modes with CMake-based build support.
- Added CUDA support for pixel-parallel rendering, GPU-friendly BVH export, and flat device-side scene buffers.
- Built validation tooling to compare CPU BVH results against brute-force intersection and CUDA BVH traversal.
- Added benchmark and progressive path tracing capabilities, including multi-sample accumulation, multi-bounce support, and live preview integration.

LinkedIn Job Matcher — Job Search Automation Tool

- Built a Streamlit-based job matching tool that compares resumes against job descriptions, detects visa/citizenship/clearance blockers with evidence snippets, and ranks opportunities using weighted skill, experience, domain, and keyword scoring.
- Implemented local SQLite persistence, saved-analysis filtering, and batch scanning workflows for LinkedIn/JobSpy results, helping prioritize higher-fit roles and reduce manual review time.

Education

MASTER OF SCIENCE IN COMPUTER SCIENCE ***Aug.2020***

DigiPen Institute of Technology

BACHELOR OF ENGINEERING IN SOFTWARE ENGINEERING ***Jun. 2015***

Nanjing University